Subjective Question Answer Evaluation System

1. Introduction:

In the modern education system, evaluating students' answers is an essential but time-consuming task for teachers and examiners. Traditional examination systems mainly focus on manual checking of subjective answers, which requires significant effort, time, and consistency from evaluators. With the increasing number of students in schools, colleges, and universities, manual evaluation becomes difficult to manage efficiently. This creates the need for an automated system that can assist in evaluating descriptive or subjective answers accurately and quickly.

The Subjective Question Answer Evaluation System is designed to automate the process of checking and grading descriptive answers using modern technologies such as Natural Language Processing (NLP) and Machine Learning. The system compares the student's answer with a predefined reference answer and evaluates the similarity, meaning, and relevance of the content. Based on this analysis, the system assigns marks automatically. This helps reduce human effort while maintaining consistency and fairness in evaluation.

The system allows teachers or administrators to create exams, add questions, and provide model answers. Students can log in to the platform and submit their responses online. After submission, the system processes the answers, compares them with the correct answers, and generates marks based on similarity algorithms. The results are stored in a database and can be viewed by both students and teachers.

Another important objective of this system is to minimize bias and errors that may occur during manual checking. Human evaluation can sometimes vary depending on the examiner's mood, workload, or interpretation of answers. An automated system follows a consistent evaluation method, ensuring fairness for all students.

In addition, the system improves efficiency in educational institutions. Teachers can save time on answer checking and focus more on teaching and improving student performance. Students also benefit from faster results and transparent evaluation. The system can be used in schools, colleges, online learning platforms, and competitive examinations.

The Subjective Question Answer Evaluation System integrates technologies such as text similarity analysis, keyword extraction, and semantic matching to assess answers more intelligently. It can also be extended with advanced Artificial Intelligence techniques in the future to improve accuracy and adaptability.

2. Scope of the system:

The scope of the Subjective Question Answer Evaluation System defines the boundaries, features, and capabilities of the proposed system. It explains what the system will accomplish, where it can be applied, and how it improves the traditional examination process.

1. Automation of Answer Evaluation:
   The primary scope of the system is to automate the evaluation of descriptive or subjective answers. Instead of manual checking by teachers, the system analyses the student's response and compares it with a predefined model answer. Using text processing techniques, the system determines similarity, relevance, and keyword matching to assign marks automatically. This reduces the time required for evaluation and helps manage examinations for a large number of students efficiently.

2. User Roles in the System:

   I. Admin: The administrator manages the entire system. Their responsibilities include:
   - Managing users (students and teachers)
   - Creating and managing exams
   - Adding subjects and questions
   - Monitoring system activity
   - Viewing reports and results

   II. Teacher/Examiner: Teachers interact with the system to conduct exams and evaluate answers. Functions include:
   - Creating subjective questions
   - Adding reference answers
   - Setting marks for questions
   - Reviewing automated evaluations
   - Generating student performance reports

3. Intelligent Answer Analysis:
   The system evaluates answers using Natural Language Processing techniques. The scope includes:
   - Keyword matching
   - Text similarity comparison
   - Tokenization and text preprocessing
   - Basic semantic analysis

4. Result Generation and Performance Analysis:
   The system automatically generates results after evaluation. The scope includes:
   - Instant marks calculation
   - Result storage in the database
   - Performance reports
   - Student progress tracking

5. Reduction of Manual Work:
   One of the main scopes of this project is reducing the workload on teachers. Traditional subjective answer checking requires significant time and effort. The system speeds up this process and improves productivity within educational institutions.

6. Database Management:
   All information is securely stored in the database, including:
   - Student records
   - Question banks
   - Model answers
   - Exam details
   - Submitted responses
   - Evaluation results

3. Proposed System:

The proposed system is a web-based application designed to automatically evaluate subjective answers written by students. Traditional evaluation of descriptive answers requires a large amount of time and effort from teachers, and it may sometimes lead to inconsistent marking. The proposed system uses Natural Language Processing (NLP) techniques and machine learning algorithms to analyse student answers and compare them with predefined model answers.

The system is developed using Python with the Django framework for backend development, HTML/CSS/JavaScript for frontend design, and a database such as SQLite or MySQL for storing questions, answers, and results.

The system aims to assist teachers by reducing manual effort, improving evaluation speed, and maintaining consistency in grading.

Advantages of the Proposed System:

- Saves time for teachers.
- Reduces manual checking.
- Provides quick results.
- Maintains consistency in evaluation.
- Easy data management.
- Accessible from anywhere.
- Reduces human errors

Limitations of the Proposed System:

- Accuracy depends on the quality of the model answer.
- Complex answers may be difficult to evaluate perfectly.
- Requires proper training of NLP models.
- Internet access is required.

4. Objectives Of System:

1. Automating the Evaluation Process:
   One of the primary objectives of the system is to automate the evaluation of subjective or descriptive answers. Traditionally, teachers spend a significant amount of time reading and grading written responses. The proposed system uses Natural Language Processing (NLP) techniques to analyse the content of students' answers and compare them with predefined model answers. By automating this process, the system reduces manual work and speeds up the evaluation procedure.

2. Reducing Teachers Workload:
   Manual checking of answer sheets can be time-consuming, especially when there are large numbers of students. The system is designed to assist teachers by automatically evaluating answers and generating marks. Teachers only need to review results if necessary. This reduces their workload and allows them to focus more on teaching and student development.

3. Providing Fast and Instant Results:
   The system aims to provide immediate evaluation results after students submit their answers. Instead of waiting days or weeks for results, students and teachers can quickly access scores and feedback. This improves the efficiency of the examination process and enhances the learning experience.

4. Managing Examination Data Efficiently:
   Another objective of the system is to store and manage examination-related data effectively. The system maintains records of:
   - Students
   - Questions
   - Model answers
   - Student responses
   - Marks and results

   This organized data storage makes it easier for teachers and administrators to access and analyse information when required.

5. Saving Time and Resources:
   The system reduces the need for paper-based examinations and manual checking. This saves time, reduces administrative work, and contributes to an environmentally friendly digital examination system.

6. Supporting Online Examination Systems:
   With the increasing use of online learning platforms, there is a need for automated assessment tools. The proposed system supports online examinations where students can log in, answer questions, and receive results through a web interface.

5. Literature Review:

1. **"Automated grading using natural language processing and semantic analysis - ScienceDirect":** The rapid growth of digital education and online examinations has increased the need for automated systems that can evaluate subjective answers efficiently. Traditional manual evaluation methods are time-consuming, inconsistent, and difficult to scale when a large number of students are involved. Researchers have therefore explored automated grading systems that use Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning to analyse students' answers and assign marks automatically. These systems aim to provide faster results, reduce human bias, and improve the overall examination process.

2. **"An automated essay scoring systems: a systematic literature review - PubMed":** Automated essay scoring and short-answer grading have been studied for several decades, but evaluating descriptive answers remains challenging because answers can be written in different ways while conveying the same meaning. Many research studies focus on understanding semantic similarity between student responses and model answers to generate fair evaluations.

6. Requirement Analysis:

1. Purpose of the System:
   The main purpose of the system is to develop an online platform that can automatically evaluate subjective answers submitted by students. The system compares student responses with model answers using text analysis techniques and generates marks accordingly.
   The system will:
   - Conduct online subjective examinations
   - Evaluate answers automatically
   - Generate results instantly
   - Maintain records of students and exams
   - Provide feedback to students and teachers

2. Functional Requirements:
   Functional requirements describe what the system should do:
   2.1 User Management: The system must allow users to register, log in, and manage their profiles. Features:
   - User registration
   - Login authentication
   - Password management
   - Role-based access control

   2.2 Question Management: Teachers and administrators must be able to manage subjective questions. Features:
   - Add questions
   - Edit or delete questions
   - Add model answers
   - Assign marks to questions

   2.3 Exam Management: The system should allow teachers to create and manage exams. Features:
   - Create exams
   - Assign questions to exams
   - Set exam duration
   - Schedule exam sessions

   2.4 Answer Submission: Students must be able to attend exams and submit answers. Features:
   - Display questions
   - Input answer text
   - Submit responses
   - Save answers in the database

   2.5 Automated Evaluation: The system must analyse the submitted answers and evaluate them. Features:
   - Compare answers with model answers
   - Keyword analysis
   - Text similarity calculation
   - Score generation

2.6 Result Management: The system should generate and display results.
Features:

- Automatic mark calculation
- Result storage
- Student result display
- Teacher review option

2.7 Reporting System: The system must provide reports and analysis.
Features:

- Student performance reports
- Exam summaries
- Result analytics
- Downloadable reports

3. Non-Functional Requirements:
Non-functional requirements describe how the system should perform:
3.1 Performance Requirements:
    - The system should handle multiple users simultaneously.
    - Evaluation should be completed quickly after answer submission.
    - The system should provide fast response time.

3.2 Security Requirements:

- Secure login authentication
- Data protection and privacy
- Role-based access control
- Secure database storage

3.3 Usability Requirements:

- Simple and user-friendly interface
- Easy navigation
- Clear instructions for exams
- Accessible to users with basic computer knowledge

3.4 Reliability Requirements:

- The system should operate without frequent errors.
- Data should be stored safely without loss.
- System backups should be possible.

3.5 Scalability Requirements:

- The system should support a large number of users.
- It should allow addition of more courses and exams in the future.

4. Hardware Requirements:
   The system requires the following hardware components:
   Server:
   - Processor: Intel i3 or higher
   - RAM: Minimum 4 GB
   - Storage: 500 GB or more

   Client:

   - Computer or smartphone
   - Internet connection

5. Software Requirements:
   The system requires the following software components:
   - **Operating System**: Windows
   - **Technologies Used:** Python, Django Framework, HTML, CSS, JavaScript
   - **Database:** SQLite
   - **Web Browser:** Chrome, Firefox, Edge

7. Environment Setup:

1. Installation of Python:
   Step 1: Download Python
   Visit the official Python website and download the latest stable version.

   Step 2: Install Python
   Run the installer and select the option Add Python to PATH before installing.

   Step 3: Verify Installation
   Open the command prompt or terminal and type:
   >> python –version

2. Installing Django Framework:
   After activating the virtual environment, install Django.
   >> pip install Django

   Check Django installation:
   >> django-admin –version
   This confirms that Django has been installed successfully.

3. Creating the Application Module:
   In Django, functionality is divided into applications. Create an application for answer evaluation.
   >> python manage.py startapp evaluation

   Add the application to settings.py
   ```
   INSTALLED_APPS = [
      'evaluation',
   ]
   ```
   This app will handle exam management, questions, and answer evaluation.

4. Database Configuration:
   Django automatically uses SQLite as the default database.
   Database settings in settings.py
   ```
   DATABASES = {
    'default': {
       'ENGINE': 'django.db.backends.sqlite3',
       'NAME': BASE_DIR / 'db.sqlite3',
    }
   }
   ```

Create database tables by running migrations.
>> python manage.py makemigrations
>> python manage.py migrate

The database is now ready to store system data.

5. Running the Application:
   Activate virtual environment.
   Start the Django server.
   >> python manage.py runserver

   Access the system through the browser.
   http://127.0.0.1:8000

   The application will now run locally.

8. Initial Development:

1. User Authentication Development:
   Authentication is essential for controlling access to the system.
   Features Implemented:
   - User registration
   - Login system
   - Logout functionality
   - Role-based access

2. Admin Panel Configuration:
   Django provides a built-in admin interface which simplifies system management.
   Admin Functionalities:
   - Add and manage users
   - Add subjects and exams
   - Manage questions and answers
   - Monitor student submissions

3. User Interface Development:
   The initial user interface is developed to allow users to interact with the system.
   Main Pages Created:
   - Home Page
   - User Registration Page
   - Login Page
   - Dashboard
   - Exam Page
   - Result Page

4. Question Management Module:
   This module allows teachers to manage questions used in examinations.
   Features:
   - Add new questions
   - Edit questions
   - Delete questions
   - Provide model answers
   - Assign marks

   All questions are stored in the database and retrieved during exams.

5. Exam Management Module:
   Teachers create and manage exams through this module.
   Functionalities:
   - Create exam
   - Select questions
   - Set exam duration
   - Schedule exam

   Students can view available exams through their dashboard.

6. Answer Submission System:
   Students submit descriptive answers through an online interface.
   Process:
   - Student logs into the system
   - Selects an available exam
   - Views questions
   - Types answers in text fields
   - Submits responses

   Answers are saved in the database for evaluation.

7. Student Exam Module:
   This module allows students to attend exams online. Students can view questions and submit descriptive answers through the system.
   Functions:
   - Display exam questions
   - Answer submission
   - Time tracking
   - Auto-save responses

8. Result and Feedback Module:
   This module generates results after evaluation and provides feedback to students. Teachers can also review the marks assigned by the system.
   Functions:
   - Result calculation
   - Performance analysis
   - Feedback display
   - Result history

9. Report and Analytics Module:
   This module provides statistical insights about exams and student performance. It helps teachers and administrators analyse results.
   Functions
   - Student performance reports
   - Exam statistics
   - Result summaries
   - Downloadable reports